

HG4052 · Week 6 practical, stage 1: the ice-cream trellis

Jurafsky & Martin, Appendix A, Fig. A.2 (the example is Eisner's). Work in pairs; keep the sheet, the notebook checks it.

The HMM

Hidden states: the weather each day, **H**ot or **C**old. Observations: how many ice creams were eaten that day, 1, 2 or 3. The diary reads **3 1 3** (day 1, day 2, day 3).

start	P(H) = 0.8	P(C) = 0.2	
transitions	H→H 0.6 H→C 0.4	C→C 0.5 C→H 0.5	
emissions	P(1 H) = 0.2 P(2 H) = 0.4 P(3 H) = 0.4	P(1 C) = 0.5 P(2 C) = 0.4 P(3 C) = 0.1	

The trellis: fill every cell, draw an arrow into each cell from the winning previous cell

	day 1: ate 3	day 2: ate 1	day 3: ate 3
H	$v(H, 1) = \text{start} \times P(3 H)$	$v(H, 2) = P(1 H) \times \max(v(H, 1) \times H \rightarrow H, v(C, 1) \times C \rightarrow H)$	$v(H, 3) = P(3 H) \times \max(\dots)$
C	$v(C, 1) = \text{start} \times P(3 C)$	$v(C, 2) = P(1 C) \times \max(v(H, 1) \times H \rightarrow C, v(C, 1) \times C \rightarrow C)$	$v(C, 3) = P(3 C) \times \max(\dots)$

Larger of the two day-3 cells: _____ Follow the arrows back. Best weather sequence: ____ ____ ____
Its probability: _____

Hints. 1. Day 1 has no previous cell: start × emission. 2. Every later cell: emission of that day's number × the larger of the two (previous cell × transition) products; the winner is the arrow. 3. Day 3: take the larger cell and follow the arrows back to day 1. The numbers shrink every day; that is normal.

Now relabel

H = oral vowel, C = nasal vowel; 1, 2, 3 = low, mid, high nasal murmur in the frame. The observation sequence 3 1 3 is now three frames of a recording. Which cells, arrows or numbers change?

In one sentence, what did the machinery need to know about weather, or about vowels, to decode?

Runner-up check: the all-hot sequence H H H has probability $0.8 \times 0.4 \times 0.6 \times 0.2 \times 0.6 \times 0.4 = 0.0092$. Is your best path better than that?